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Article in *Applied Numerical Mathematics* · May 2019

DOI: 10.1016/j.apnum.2019.05.012

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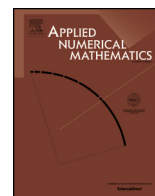


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A modified conjugate gradient method for monotone nonlinear equations with convex constraints

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ARTICLE INFO

Article history:

Received 1 August 2018

Received in revised form 1 May 2019

Accepted 9 May 2019

Available online xxxx

Keywords:

Spectral gradient method

Nonlinear monotone equations

Projection method

Compressive sensing

ABSTRACT

In this paper, a modified Hestenes-Stiefel (HS) spectral conjugate gradient (CG) method for monotone nonlinear equations with convex constraints is proposed based on projection technique. The method can be viewed as an extension of a modified HS-CG method for unconstrained optimization proposed by Amini et al. (Optimization Methods and Software, pp: 1-13, 2018). A new search direction is obtained by incorporating the idea of spectral gradient parameter and some modification of the conjugate gradient parameter. The proposed method is derivative-free and requires low memory which makes it suitable for large scale monotone nonlinear equations. Global convergence of the method is established under suitable assumptions. Preliminary numerical comparisons with some existing methods are given to show the efficiency of our proposed method. Furthermore, the proposed method is successfully applied to solve sparse signal reconstruction in compressive sensing.

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1. Introduction

In this paper, we consider the constrained monotone nonlinear equations of the form

$$F(x) = 0, \quad x \in \Omega, \quad (1.1)$$

where $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous and monotone. The monotonicity of the mapping F means that $(F(x) - F(y))^T(x - y) \geq 0$, $\forall x, y \in \mathbb{R}^n$. The feasible set $\Omega \subset \mathbb{R}^n$ is a nonempty closed and convex and $\mathbb{R}^n$ is the n -dimensional Euclidean space. Under these conditions, the solution set of problem (1.1) is convex [21]. Problem (1.1) has many practical applications. For example, it appears as a subproblem in generalized proximal algorithms with Bregman distance [11]. Moreover, some

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<https://doi.org/10.1016/j.apnum.2019.05.012>

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monotone variational inequality problems can be converted into systems of monotone equations [37]. Furthermore, l_1 -norm regularized optimization problems can be reformulated as monotone nonlinear equations [9].

There are many iterative methods for solving the unconstrained version of (1.1), in which $\Omega = \mathbb{R}^n$. For instance, we can cite the Newton Method [13,26,38], the Fixed Newton Method [13], quasi-Newton methods [7,16,26], Gauss-Newton method, Levenberg-Marquart [29] and their variants. With a good starting point, these methods converge very fast which make them very attractive. However, they are not suitable to handle large-scale monotone nonlinear equations because they need to solve a system of linear equations in each iteration using the Jacobian matrix of $F(x)$ or its approximation. In recent years, the problem (1.1) has received much attention from researchers. Methods such as trust-region methods, quasi-Newton methods, conjugate gradient methods, and so on, are used to solve this type of problem. Notwithstanding, conjugate gradient methods are sometimes the usual choice, due to their simplicity, global convergence, and low memory requirement. There are conjugate gradient methods and spectral gradient methods for solving unconstrained optimization that have been extended to solve system of nonlinear equations (see, [1,2,4,20,32–34]).

Bellavia and Morini [6] introduced subspace trust-region method for large bound-constrained nonlinear systems. Their idea was to investigate the trust-region problem in a small subspace while global and local convergence are still attained. The global and superlinear/quadratic convergence of the method were shown under standard assumptions. In [12], Kanzow and Klug proposed an affine-scaling trust-region method for solving semismooth system of equations with box constraints. This method can be applied to both continuously differentiable and semismooth systems of equations. They have shown strong global and local convergence properties under suitable assumptions. A quasi-Newton approach for solving constrained nonlinear system of equations was proposed by Marini et al. [19], where they embedded a quasi-Newton step into a non-monotone line search strategy based on the projection map. The numerical results presented show their method is very efficient.

The projection technique proposed by Solodov and Svaiter [23] has motivated some researchers to further develop methods for solving monotone nonlinear equations with convex constraints. For example, Wang et al. [24] extended the work of Solodov and Svaiter [23] to solve convex constrained monotone equations. The main feature of their method is that, a well-conditioned system of linear equations is needed to be solved at each step. After initialization, the system of linear equations is solved approximately to get a trial point and then use projection technique to obtain the next iterate. They have shown global convergence and linear convergence rate under some suitable conditions. To improve the convergence rate of the projection method in [24], Wang and Wang [25] modified it and proposed another projection method for solving constrained system of nonlinear equations with superlinear convergence rate. Yu et al. [31] extended the spectral gradient method proposed by Zhang and Zhou [36] to solve monotone nonlinear equations with convex constraints which do not require solving linear system of equations. In [18], Liu and Li modified the work of Xiao and Zhu [27] and proposed a projection method for solving monotone nonlinear equations with convex constraints by replacing the CG parameter β_k^{CGD} with a modified one denoted by β_k^{PCG} . The method can also be viewed as an extension of the well-known CG_DESCENT method for solving unconstrained optimization problem proposed by Hager and Zhang [10]. Very recently, inspired by the projection technique in [23], Yuan [35] extended the well-known CG_DESCENT method to solve problem (1.1) and also proved the global convergence of the method under some appropriate conditions.

Inspired by the above contributions, we extend the modified Hestenes-Stiefel (HS) conjugate gradient (CG) method proposed in [3] to solve problem (1.1) by combining the spectral gradient parameter defined in [5] and a new CG parameter to define the search direction of our algorithm. Under some suitable assumptions, we show that the direction satisfies the sufficient descent condition and we establish the global convergence of the method. We present some preliminary numerical experiments to illustrate the efficiency of our proposed method. In addition, we apply our proposed method to recover a sparse signal arising in compressive sensing.

The remaining part of this paper is organized as follows. In section 2, we describe our proposed method and its algorithm. We establish the global convergence in section 3. In section 4, we report numerical experiments to show the efficiency of our method. In section 5, we apply the proposed algorithm to deal with the sparse signal reconstruction in compressive sensing and we give our conclusion in section 6.

2. Motivation and proposed method

In this section, we give some preliminaries and then recall the modified HS-CG method proposed by Amini et al. [3]. Throughout this paper, $\|\cdot\|$ denotes the Euclidean norm. Let $\Omega \subset \mathbb{R}^n$ be a nonempty closed and convex set. For any $x \in \mathbb{R}^n$, its projection onto Ω is defined as $P_\Omega(x) = \arg\min\{\|x - y\| : y \in \Omega\}$.

The mapping $P_\Omega : \mathbb{R}^n \rightarrow \Omega$ is called a projection operator. An interesting property is that this operator $P_\Omega(\cdot)$ is nonexpansive, i.e.

$$\|P_\Omega(x) - P_\Omega(y)\| \leq \|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

Consequently, we have

$$\|P_\Omega(x) - y\| \leq \|x - y\|, \quad \forall y \in \Omega. \quad (2.1)$$

On the other hand, let us consider the modified HS-CG method proposed by Amini et al. [3] to minimize $f(x)$, where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuously differentiable function. Starting from an initial guess x_0 , the method generates a sequence $\{x_k\}$ using the following formula:

$$x_{k+1} = x_k + \alpha_k d_k, \quad (2.2)$$

where $\alpha_k > 0$ is the step length obtained via some suitable line search and the search direction d_k is defined by

$$d_k = \begin{cases} -g_k, & \text{if } k = 0, \\ -g_k + \bar{\beta}_k'^{MHS+} d_{k-1}, & \text{if } k > 0, \end{cases} \quad (2.3)$$

where $g_k = \nabla f(x_k)$, $\bar{\beta}_k'^{MHS+} = \max\{\bar{\beta}_k'^{MHS}, 0\}$.

Here,

$$\bar{\beta}_k'^{MHS} = \frac{g_k^T y'_{k-1}}{y_{k-1}'^T d_{k-1}} \theta_k - \gamma \left(\frac{\|y'_{k-1}\| \theta_k}{y_{k-1}'^T d_{k-1}} \right)^2 g_k^T d_{k-1}, \quad (2.4)$$

where $y'_{k-1} = g_k - g_{k-1}$, $\theta_k' = 1 - (g_k^T d_{k-1})^2 / \|g_k\|^2 \|d_{k-1}\|^2$, and $\gamma > 1/4$.

It was shown that, if $y_{k-1}'^T d_{k-1} \neq 0$, the search direction d_k satisfies the sufficient descent condition $g_k^T d_k \leq \bar{c} \|g_k\|^2$, $\forall k$, where $\bar{c}$ is a positive constant. This property is very crucial for an iterative method to be globally convergent. This makes the modified HS-CG method very useful.

Motivated by the work of Xiao and Zhu [27], we propose a modified method for solving problem (1.1) and define the search direction as follows:

$$d_k = \begin{cases} -F_k, & \text{if } k = 0, \\ -\lambda_k F_k + \bar{\beta}_k'^{MHS+} d_{k-1}, & \text{if } k > 0, \end{cases} \quad (2.5)$$

where

$$\bar{\beta}_k'^{MHS+} = \max\{\bar{\beta}_k'^{MHS}, 0\}, \quad (2.6)$$

$$\bar{\beta}_k'^{MHS} = \frac{\|s_{k-1}\|^2 F_k^T y_{k-1}}{(s_{k-1}^T y_{k-1})(y_{k-1}'^T d_{k-1})} \theta_k - \gamma \left(\frac{\|s_{k-1}\| \|y_{k-1}\| \theta_k}{y_{k-1}'^T d_{k-1}} \right)^2 \frac{F_k^T d_{k-1}}{s_{k-1}^T y_{k-1}} \quad (2.7)$$

$$\theta_k = 1 - (F_k^T d_{k-1})^2 / \|F_k\|^2 \|d_{k-1}\|^2, \quad (2.8)$$

$$\lambda_k = \frac{s_{k-1}^T s_{k-1}}{s_{k-1}^T y_{k-1}}, \quad (2.9)$$

and F_k denotes the function evaluation of F at x_k , $y_{k-1} = F_k - F_{k-1} + \eta s_{k-1}$, $s_{k-1} = x_k - x_{k-1}$, $\eta > 0$. This definition of y_{k-1} is chosen to ensure $s_{k-1}^T y_{k-1} > 0$, (see, Remark (a)).

Next, we state the algorithm of the proposed method which we call a modified Hestenes-Stiefel spectral conjugate gradient projection (MHSP) method for solving problem (1.1).

Algorithm 1 (MHSP).

Step 0. Given $x_0 \in \mathbb{R}^n$, $\rho \in (0, 1)$, $\sigma, \kappa > 0$, $\eta \geq 0$, stopping tolerance $\varepsilon \geq 0$. Set $k = 0$.

Step 1. Compute F_k . If $\|F_k\| \leq \varepsilon$, stop.

Step 2. Compute d_k using Equations (2.5)–(2.9).

Step 3. Determine $\alpha_k = \max\{\kappa \rho^i : i = 0, 1, 2, \dots\}$ such that

$$-F(x_k + \alpha_k d_k)^T d_k \geq \sigma \alpha_k \|d_k\|^2. \quad (2.10)$$

Step 4. Set $z_k = x_k + \alpha_k d_k$. If $z_k \in \Omega$ and $\|F(z_k)\| \leq \varepsilon$, stop. Otherwise, compute the next iterate by

$$x_{k+1} = P_\Omega [x_k - \mu_k F(z_k)], \text{ where } \mu_k = \frac{F(z_k)^T (x_k - z_k)}{\|F(z_k)\|^2}. \quad (2.11)$$

Step 5. Set $k := k + 1$ and go to step 1.

For the purpose of this paper, we need to make the following assumptions:

(Ai) The mapping $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is Lipschitz continuous, i.e., there exists a positive constant L such that

$$\|F(x) - F(y)\| \leq L \|x - y\|, \quad \forall x, y \in \mathbb{R}^n. \quad (2.12)$$

(Aii) The mapping $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is uniformly monotone, i.e., there exists a positive real number c such that

$$(F(x) - F(y))^T(x - y) \geq c\|x - y\|^2, \quad \forall x, y \in \mathbb{R}^n. \quad (2.13)$$

(Aiii) The solution set of (1.1) is nonempty and is denoted by $\overline{X}$.

Next, we give the following remarks:

Remarks.

(a) By the definition of y_{k-1} and the monotonicity of F , we have

$$s_{k-1}^T y_{k-1} = s_{k-1}^T (F_k - F_{k-1}) + \eta s_{k-1}^T s_{k-1} \geq \eta \|s_{k-1}\|^2 > 0. \quad (2.14)$$

This means λ_k defined by (2.9) is well defined.

In addition, since (2.14) holds, then by (2.12) we have

$$y_{k-1}^T s_{k-1} = (F_k - F_{k-1})^T (x_k - x_{k-1}) + \eta \|s_{k-1}\|^2 \leq (L + \eta) \|s_{k-1}\|^2. \quad (2.15)$$

From Equations (2.14) and (2.15), we have $(L + \eta) \|s_{k-1}\|^2 \geq y_{k-1}^T s_{k-1} \geq \eta \|s_{k-1}\|^2$, which implies $\frac{\|s_{k-1}\|^2}{s_{k-1}^T y_{k-1}} \leq \frac{1}{\eta}$, and $\frac{\|s_{k-1}\|^2}{s_{k-1}^T y_{k-1}} \geq \frac{1}{L + \eta} = q$. Therefore, we have

$$q \leq \lambda_k \leq \frac{1}{\eta}. \quad (2.16)$$

(b) Assumption (Ai) implies

$$\|y_{k-1}\| = \|F_k - F_{k-1} + \eta s_{k-1}\| \leq (L + \eta) \|s_{k-1}\| = (L + \eta) \alpha_{k-1} \|d_{k-1}\|. \quad (2.17)$$

(c) By assumption (Aii), we obtain

$$y_{k-1}^T d_{k-1} = (F_k - F_{k-1} + \eta s_{k-1})^T \frac{s_{k-1}}{\alpha_{k-1}} \geq (c + \eta) \frac{\|s_{k-1}\|^2}{\alpha_{k-1}} = (c + \eta) \alpha_{k-1} \|d_{k-1}\|^2, \quad (2.18)$$

which guarantees that $y_{k-1}^T d_{k-1} > 0$ provided the solution is not yet achieved. Thus, the CG parameter $\overline{\beta}_k^{MHS+}$ is well defined.

(d) By Cauchy-Schwarz inequality, $|F_k^T d_{k-1}| \leq \|F_k\| \|d_{k-1}\|$, and therefore θ_k defined in (2.8) satisfies $0 \leq \theta_k \leq 1$.

2.1. Related works

In this subsection, we briefly compare our proposed method with some existing methods and explain their differences.

- (a) If $\overline{\beta}_k^{MHS+} = 0$, then the MHSP method automatically reduces to the spectral gradient method by Zhang and Zhou [36].
- (b) The main difference between the new search direction d_k defined by (2.5)–(2.9) and the one defined by (2.3)–(2.4) is the spectral gradient parameter λ_k and the modified CG parameter $\overline{\beta}_k^{MHS}$.
- (c) One major difference between our proposed MHSP method and the PCG method in [18] lies in the definition of the search direction d_k and the choice of the CG parameter β_k . The proposed MHSP algorithm uses both spectral parameter λ_k and CG parameter $\overline{\beta}_k^{MHS+}$ to compute its search direction d_k while PCG algorithm uses only CG parameter β_k^{PCG} to compute its search direction. In addition, we can see from Equation (2.6) that whenever the CG parameter $\overline{\beta}_k^{MHS} < 0$, MHSP algorithm uses spectral gradient method to compute the search direction d_k which is not the case with PCG algorithm.

3. Convergence analysis

In this section, we establish the global convergence of our proposed method. We begin with the following Lemmas which will be useful in our analysis.

Lemma 3.1. [18] Suppose the sequence $\{x_k\}$ and $\{z_k\}$ are generated by Algorithm 1. Let the assumptions (Ai)–(Aiii) hold, then the sequences $\{x_k\}$, and $\{z_k\}$ are bounded. In addition, we have

$$\lim_{k \rightarrow \infty} \|x_k - z_k\| = 0, \quad (3.1)$$

and

$$\lim_{k \rightarrow \infty} \|x_{k+1} - x_k\| = 0. \quad (3.2)$$

From Lemma 3.1, we can deduce that there exists a positive constant ω for which

$$\|F(x_k)\| \leq \omega, \quad \forall k \geq 0. \quad (3.3)$$

Lemma 3.2. The search direction d_k defined by (2.5)–(2.9) satisfies the descent property $F_k^T d_k \leq -\Gamma \|F_k\|^2$ for all $k \geq 0$ where Γ is a positive constant.

Proof. For $k = 0$, from (2.5) we have $F_0^T d_0 = -\|F_0\|^2$, i.e. $\Gamma = 1$. Now, for $k > 0$ and $\bar{\beta}_k^{MHS+} = 0$ by (2.5) and (2.16), we have

$$\begin{aligned} F_k^T d_k &= -\lambda_k \|F_k\|^2 \\ &\leq -q \|F_k\|^2. \end{aligned}$$

Since q is a positive constant, we are done. On the other hand, for $k > 0$ and $\bar{\beta}_k^{MHS+} = \bar{\beta}_k^{MHS} > 0$, we have the following from (2.5)–(2.9)

$$\begin{aligned} F_k^T d_k &= -\lambda_k \|F_k\|^2 + \bar{\beta}_k^{MHS} F_k^T d_{k-1} \\ &= -\frac{s_{k-1}^T s_{k-1}}{s_{k-1}^T y_{k-1}} \|F_k\|^2 + \left[\frac{\|s_{k-1}\|^2 F_k^T y_{k-1}}{(s_{k-1}^T y_{k-1})(y_{k-1}^T d_{k-1})} \theta_k - \gamma \left(\frac{\|s_{k-1}\| \|y_{k-1}\| \theta_k}{y_{k-1}^T d_{k-1}} \right)^2 \frac{F_k^T d_{k-1}}{s_{k-1}^T y_{k-1}} \right] F_k^T d_{k-1} \\ &= -\frac{s_{k-1}^T s_{k-1}}{s_{k-1}^T y_{k-1}} \|F_k\|^2 + \frac{\|s_{k-1}\|^2}{s_{k-1}^T y_{k-1}} \left[\frac{F_k^T y_{k-1}}{y_{k-1}^T d_{k-1}} \theta_k - \gamma \left(\frac{\|y_{k-1}\| \theta_k}{y_{k-1}^T d_{k-1}} \right)^2 F_k^T d_{k-1} \right] F_k^T d_{k-1} \\ &= \frac{s_{k-1}^T s_{k-1}}{s_{k-1}^T y_{k-1}} \left[\frac{(y_{k-1}^T F_k)(y_{k-1}^T d_{k-1})(F_k^T d_{k-1}) \theta_k - \|F_k\|^2 (y_{k-1}^T d_{k-1})^2 - \gamma \|y_{k-1}\|^2 (F_k^T d_{k-1})^2 \theta_k^2}{(y_{k-1}^T d_{k-1})^2} \right]. \end{aligned}$$

Now, applying the inequality $u^T v \leq \frac{1}{2}(\|u\|^2 + \|v\|^2)$ to the first term of the last equality with

$$u = \frac{1}{\sqrt{2\gamma}} (y_{k-1}^T d_{k-1}) F_k \quad \text{and} \quad v = \sqrt{2\gamma} (F_k^T d_{k-1}) y_{k-1} \theta_k,$$

we obtain

$$\begin{aligned} F_k^T d_k &\leq -\lambda_k \left(1 - \frac{1}{4\gamma} \right) \|F_k\|^2 \\ &\leq -q \left(1 - \frac{1}{4\gamma} \right) \|F_k\|^2. \end{aligned}$$

The last inequality follows from the inequality (2.16). Therefore, since $q > 0$, we let $\Gamma = q \left(1 - \frac{1}{4\gamma} \right)$ where γ satisfies the same condition in (2.4), and we have

$$F_k^T d_k \leq -\Gamma \|F_k\|^2. \quad \square \quad (3.4)$$

Lemma 3.3. Let $\{d_k\}$ be the sequence of directions generated by Algorithm 1, then there exists a constant $M > 0$ such that $\|d_k\| \leq M$, $\forall k = 0, 1, 2, \dots$.

Proof. For $k = 0$, we have from (2.5) and (3.3), $\|d_0\| = \|F_0\| \leq \omega$.

Again, for $k > 0$ and $\bar{\beta}_k^{MHS+} = 0$, we have by (2.5), (2.16) and (3.3)

$$\begin{aligned} \|d_k\| &= \|-\lambda_k F_k\| \\ &= \lambda_k \|F_k\| \\ &\leq \frac{\omega}{\eta}. \end{aligned}$$

Since η and ω are positive constants, the result holds. On the other hand, for $k > 0$ and $\bar{\beta}_k^{MHS+} = \bar{\beta}_k^{MHS} > 0$, by Equations (2.5)–(2.9) and remarks (a)–(d) we have

$$\begin{aligned}
 \|d_k\| &= \left\| -\lambda_k F_k + \bar{\beta}_k^{MHS} d_{k-1} \right\| \\
 &\leq \lambda_k \|F_k\| + \left| \bar{\beta}_k^{MHS} \right| \|d_{k-1}\| \\
 &\leq \lambda_k \|F_k\| + \left[\frac{\|s_{k-1}\|^2 |F_k^T y_{k-1}|}{(s_{k-1}^T y_{k-1})(y_{k-1}^T d_{k-1})} \theta_k + \gamma \left(\frac{\|s_{k-1}\| \|y_{k-1}\| \theta_k}{y_{k-1}^T d_{k-1}} \right)^2 \frac{|F_k^T d_{k-1}|}{s_{k-1}^T y_{k-1}} \right] \|d_{k-1}\| \\
 &\leq \lambda_k \|F_k\| + \frac{\|s_{k-1}\|^2}{s_{k-1}^T y_{k-1}} \left[\frac{\|F_k\| \|y_{k-1}\|}{y_{k-1}^T d_{k-1}} + \frac{\gamma \|y_{k-1}\|^2}{(y_{k-1}^T d_{k-1})^2} \|F_k\| \|d_{k-1}\| \right] \|d_{k-1}\| \\
 &\leq \lambda_k \|F_k\| + \lambda_k \left[\frac{(L + \eta) \alpha_{k-1} \|F_k\| \|d_{k-1}\|}{(c + \eta) \alpha_{k-1} \|d_{k-1}\|^2} + \frac{\gamma (L + \eta)^2 \alpha_{k-1}^2 \|d_{k-1}\|^2}{(c + \eta)^2 \alpha_{k-1}^2 \|d_{k-1}\|^4} \|F_k\| \|d_{k-1}\| \right] \|d_{k-1}\| \\
 &= \lambda_k \left[1 + \frac{(L + \eta)}{(c + \eta)} + \frac{\gamma (L + \eta)^2}{(c + \eta)^2} \right] \|F_k\| \\
 &\leq \frac{1}{\eta} \left[1 + \frac{(L + \eta)}{(c + \eta)} + \frac{\gamma (L + \eta)^2}{(c + \eta)^2} \right] \omega.
 \end{aligned}$$

Therefore if we let $M = \frac{1}{\eta} \left[1 + \frac{(L + \eta)}{(c + \eta)} + \frac{\gamma (L + \eta)^2}{(c + \eta)^2} \right] \omega$, then we have

$$\|d_k\| \leq M. \quad (3.5)$$

Hence the search direction is bounded. $\square$

The following theorem establishes the global convergence of Algorithm 1.

Theorem 3.4. Let $\{x_k\}$ be a sequence generated by Algorithm 1. Suppose that assumptions (Ai) and (Aiii) hold, then

$$\liminf_{k \rightarrow \infty} \|F(x_k)\| = 0. \quad (3.6)$$

Proof. Suppose on the contrary that (3.6) does not hold, then there exists a positive constant ϵ for which

$$\|F_k\| \geq \epsilon \quad \forall k \geq 0. \quad (3.7)$$

From Lemma 3.1 and the definition of z_k , we can deduce

$$\lim_{k \rightarrow \infty} \alpha_k \|d_k\| = 0. \quad (3.8)$$

If $\alpha_k \neq \kappa$, then by the definition of α_k , $\rho^{-1} \alpha_k$ does not satisfy the line search (2.10), i.e.,

$$-F(x_k + \rho^{-1} \alpha_k d_k)^T d_k < \sigma \rho^{-1} \alpha_k \|d_k\|^2,$$

then by (3.4), (2.12) and Cauchy-Schwarz inequality, we have

$$\begin{aligned}
 \Gamma \|F_k\|^2 &\leq -F_k^T d_k \\
 &= (F(x_k + \rho^{-1} \alpha_k d_k) - F_k)^T d_k - F(x_k + \rho^{-1} \alpha_k d_k)^T d_k \\
 &\leq L \rho^{-1} \alpha_k \|d_k\|^2 + \sigma \rho^{-1} \alpha_k \|d_k\|^2.
 \end{aligned}$$

This implies

$$\alpha_k \geq \frac{\rho \Gamma}{L + \sigma} \frac{\|F_k\|^2}{\|d_k\|^2}.$$

Then by (3.7) and (3.5) we have,

$$\alpha_k \|d_k\| \geq \frac{\rho \Gamma}{(L + \sigma)} \frac{\|F_k\|^2}{\|d_k\|} \geq \frac{\rho \Gamma}{(L + \sigma)} \frac{\epsilon^2}{M},$$

which contradicts (3.8). Thus, (3.6) holds.

Furthermore, by the continuity of F , the sequence $\{x_k\}$ has some accumulation point $\bar{x}$ such that $F(\bar{x}) = 0$. Since $\{\|x_k - \bar{x}\|\}$ converges (from Lemma 3.1) and $\bar{x}$ is an accumulation point of $\{x_k\}$, it follows that $\{x_k\}$ converges to $\bar{x}$. $\square$

Table 1

The initial points used for the test problems.

Initial Points (IP)	Values
x_1	$(1, 1, 1, \dots, 1)^T$
x_2	$(0.1, 0.1, 0.1, \dots, 0.1)^T$
x_3	$(\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \dots, \frac{1}{2^n})^T$
x_4	$(1 - \frac{1}{n}, 2 - \frac{2}{n}, 2 - \frac{3}{n}, \dots, n - 1)^T$
x_5	$(0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n})^T$
x_6	$(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n})^T$
x_7	$(\frac{n-1}{n}, \frac{n-2}{n}, \frac{n-3}{n}, \dots, 0)^T$
x_8	$(\frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots, 1)^T$

4. Numerical experiments

This section gives a performance comparison of our proposed algorithm MHSP with the PCG method [18] and the PDY method [17] for solving convex constrained nonlinear monotone equations (1.1).

- **Algorithm 1 (MHSP):** we set $\kappa = 1$, $\sigma = 0.001$, $\rho = 0.8$, $\eta = 0.01$ and $\gamma = 2$.
- **PCG :** All parameters are as in [18].
- **PDY :** All parameters are as in [17].

All codes were written in MATLAB R2017a and run on a PC with intel Core(TM) i5-8250u processor with 4 GB of RAM and CPU 1.60 GHZ. We solved eight (8) constrained test problems with eight (8) different initial points (IP) (See, Table 1). We used $n = 5,000$ for the dimension (DIM) for all problems considered except problem 8 where $n = 4$. The iteration is terminated whenever the inequality $\|F(x_k)\| \leq 10^{-6}$ or $\|F(z_k)\| \leq 10^{-6}$ is satisfied. Failure is recorded when the number of iterations exceeds 1,000 and the stopping criterion above mentioned has not been satisfied.

In Table 2, we present the results of the following information: the number of iterations (ITER) needed by each solver to converge to an approximate solution, the number of function evaluation (FVAL), the CPU time in seconds (TIME), and the norm of the objective function F at the approximate solution (NORM). The problems are denoted by P_i , $i = 1, 2, \dots, 8$. From our experiment, no solution was obtained by any of the three solvers with respect to P_1 , P_2 , and P_5 using the initial point x_4 and therefore, the information was not captured in Table 2.

We use the following test problems where $F(x) = (f_1(x), f_2(x), \dots, f_n(x))^T$, and $x = (x_1, x_2, \dots, x_n)^T$:

Problem 1. [15]

$$f_1(x) = e^{x_1} - 1$$

$$f_i(x) = e^{x_i} + x_{i-1} - 1, \quad i = 1, 2, \dots, n-1,$$

where $\Omega = \mathbb{R}_+^n$.

Problem 2. [15]

$$f_i(x_i) = \log(x_i + 1) - \frac{x_i}{n}, \quad i = 1, 2, \dots, n,$$

where $\Omega = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i \leq n, x_i > -1, i = 1, 2, \dots, n\}$.

Problem 3. [14]

$$f_i(x) = 2x_i - \sin |x_i|, \quad i = 1, 2, \dots, n,$$

where $\Omega = \mathbb{R}_+^n$.

Problem 4. [14]

$$f_i(x) = \min[\min(|x_i|, x_i^2), \max(|x_i|, x_i^3)], \quad i = 1, 2, \dots, n,$$

where $\Omega = \mathbb{R}_-^n$.

Table 2

Numerical results obtained by MHSP, PCG and PDY methods with the given problems.

MHSP							PCG			PDY			
PROBLEM	IP	ITER	FVAL	TIME	NORM	ITER	FVAL	TIME	NORM	ITER	FVAL	TIME	NORM
P1	x ₁	7	27	0.06078	4.32E-08	23	116	0.583728	9.41E-07	18	91	0.073096	9.38E-07
	x ₂	6	23	0.017918	1.53E-08	28	133	0.068825	5.69E-07	16	81	0.075295	7.61E-07
	x ₃	7	30	0.020366	1.12E-07	31	140	0.049236	9.87E-07	17	86	0.068846	4.7E-07
	x ₅	8	30	0.009937	2.72E-07	23	116	0.061397	8.2E-07	18	91	0.022835	5.35E-07
	x ₆	8	32	0.014269	2.29E-07	32	145	0.091912	7.16E-07	17	86	0.0265	7.33E-07
	x ₇	8	30	0.010516	2.76E-07	29	139	0.074311	4.32E-07	18	91	0.077301	5.36E-07
	x ₈	8	30	0.016517	2.74E-07	23	116	0.141639	8.2E-07	18	91	0.032226	5.35E-07
	x ₁	12	49	0.021576	6.99E-07	12	42	0.110192	4.59E-07	18	72	0.043374	5.71E-07
P2	x ₂	10	40	0.03327	9.37E-07	10	36	0.046087	9.33E-07	14	56	0.021205	5.44E-07
	x ₃	12	52	0.017261	5.33E-07	10	36	0.039016	6.76E-07	14	56	0.018223	3.76E-07
	x ₅	12	49	0.016232	9.17E-07	13	46	0.017483	2.53E-07	17	68	0.023508	9.87E-07
	x ₆	12	50	0.037866	3.46E-07	13	46	0.015933	1.04E-07	15	60	0.019651	4.2E-07
	x ₇	12	49	0.076398	9.17E-07	13	46	0.018905	2.53E-07	17	68	0.036436	9.87E-07
	x ₈	12	49	0.040815	9.17E-07	13	46	0.063082	2.54E-07	17	68	0.041159	9.87E-07
	x ₁	6	20	0.017696	8.45E-07	13	47	0.022125	1.23E-07	18	73	0.052626	5.41E-07
	x ₂	6	21	0.009108	1.63E-08	11	40	0.045364	7.85E-07	16	65	0.024504	3.74E-07
P3	x ₃	5	17	0.010838	5.62E-07	10	36	0.012699	6.81E-07	13	53	0.064406	8.83E-07
	x ₄	22	88	0.043515	2.2E-08	25	105	0.037042	2.04E-07	27	117	0.14332	4.78E-07
	x ₅	7	23	0.016172	9.27E-08	13	47	0.038519	6.76E-07	17	69	0.044956	8.68E-07
	x ₆	7	25	0.012789	3.73E-08	11	40	0.020967	5.29E-07	14	57	0.037517	7.41E-07
	x ₇	7	23	0.018231	9.27E-08	13	47	0.031735	6.76E-07	17	69	0.021261	8.68E-07
	x ₈	7	23	0.012302	9.3E-08	13	47	0.034248	6.77E-07	17	69	0.079498	8.68E-07
	x ₁	2	8	0.004691	0	2	8	0.018252	0	2	8	0.016873	0
	x ₂	2	7	0.009336	0	2	7	0.005158	0	2	7	0.00841	0
P4	x ₃	2	7	0.016445	0	2	7	0.006049	0	2	7	0.005282	0
	x ₄	2	8	0.006622	0	2	8	0.006696	0	2	8	0.014275	0
	x ₅	2	7	0.005762	0	2	7	0.006534	0	2	7	0.013029	0
	x ₆	2	7	0.013483	0	2	7	0.041723	0	2	7	0.03151	0
	x ₇	2	7	0.009845	0	2	7	0.006226	0	2	7	0.014037	0
	x ₈	2	7	0.006164	0	2	7	0.017608	0	2	7	0.005167	0
	x ₁	7	25	0.017552	9.29E-07	12	44	0.064006	3.45E-07	17	69	0.018643	6.59E-07
	x ₂	6	20	0.012855	1.18E-07	11	40	0.040277	5.1E-07	16	65	0.014934	3.86E-07
P5	x ₃	6	21	0.00596	6.7E-07	10	37	0.037015	6.78E-07	13	53	0.012861	8.83E-07
	x ₅	11	65	0.064019	0	14	52	0.011734	5.66E-07	18	73	0.016815	3.69E-07
	x ₆	8	29	0.012943	2.68E-08	13	48	0.011385	1.98E-07	15	62	0.028418	4.55E-07
	x ₇	11	65	0.043091	0	14	52	0.031733	5.66E-07	18	73	0.068782	3.69E-07
	x ₈	11	72	0.046268	0	14	52	0.024362	5.68E-07	18	73	0.019131	3.7E-07
	x ₁	11	44	0.074955	8.4E-07	13	47	0.067188	5.51E-07	18	73	0.039147	7.08E-07
	x ₂	12	47	0.063542	2.66E-07	13	47	0.063445	8.39E-07	19	77	0.059529	3.58E-07
	x ₃	12	47	0.036351	1.32E-08	13	47	0.062964	8.71E-07	19	77	0.074455	3.72E-07
P6	x ₄	8	26	0.04081	5.11E-08	18	64	0.106534	6.56E-07	25	101	0.076606	5.41E-07
	x ₅	12	47	0.060951	1.08E-08	13	47	0.026561	7.17E-07	18	73	0.066351	9.22E-07
	x ₆	12	47	0.029825	1.32E-08	13	47	0.027495	8.71E-07	19	77	0.111725	3.72E-07
	x ₇	12	47	0.066457	1.08E-08	13	47	0.073386	7.17E-07	18	73	0.10258	9.22E-07
	x ₈	12	47	0.063222	1.08E-08	13	47	0.045284	7.17E-07	18	73	0.0474	9.22E-07
	x ₁	6	23	0.032564	3.2E-07	11	51	0.038281	6.77E-07	7	29	0.011212	6.1E-08
	x ₂	12	50	0.053809	4.72E-07	11	50	0.043915	5.02E-07	18	90	0.051343	5.59E-07
	x ₃	12	51	0.040694	6.07E-07	12	55	0.02413	7.36E-08	18	90	0.080323	8.22E-07
P7	x ₄	18	76	0.065829	2.63E-07	25	109	0.063429	2.12E-07	27	131	0.078603	5.3E-07
	x ₅	12	48	0.017395	8.3E-09	15	69	0.027536	9.08E-07	19	95	0.057435	9.15E-07
	x ₆	14	66	0.044164	1.73E-08	13	59	0.068347	1.32E-07	18	90	0.027113	8.22E-07
	x ₇	12	48	0.018183	8.3E-09	15	69	0.049742	9.08E-07	19	95	0.057441	9.15E-07
	x ₈	12	47	0.039429	8.32E-09	15	69	0.044314	9.13E-07	19	95	0.030057	9.16E-07
	x ₁	60	463	0.405101	9.91E-07	123	922	0.227072	9.78E-07	71	566	0.052647	9.33E-07
	x ₂	79	364	0.077314	8.88E-07	123	921	0.0586	9.73E-07	71	567	0.064274	8.08E-07
	x ₃	132	557	0.058722	9.49E-07	116	871	0.064195	8.95E-07	66	527	0.051494	8.47E-07
P8	x ₄	42	345	0.025753	5.3E-07	124	928	0.039037	9.72E-07	74	588	0.036896	9.82E-07
	x ₅	58	450	0.036494	7.95E-07	118	884	0.067321	9.98E-07	73	582	0.052605	9.27E-07
	x ₆	54	422	0.057183	6.25E-07	123	922	0.022624	9.41E-07	69	550	0.031929	8.62E-07
	x ₇	34	273	0.009997	2.96E-07	74	555	0.044363	8.25E-07	55	439	0.037217	8.54E-07
	x ₈	56	429	0.026794	8.14E-07	119	892	0.055025	9.97E-07	73	583	0.053655	8.9E-07

Problem 5. [39]

$$f_i(x) = e^{x_i} - 1, \quad i = 1, 2, \dots, n,$$

where $\Omega = \mathbb{R}_+^n$.

Problem 6. [14]

$$f_1(x) = x_1 - e^{\cos(h(x_1+x_2))}$$

$$f_i(x) = x_i - e^{\cos(h(x_{i-1}+x_i+x_{i+1}))}, \quad i = 1, 2, \dots, n-1,$$

$$f_n(x) = x_n - e^{\cos(h(x_{n-1}+x_n))},$$

where $h = \frac{1}{n+1}$ and $\Omega = \mathbb{R}_+^n$.

Problem 7. [30]

$$f_i(x) = x_i - \sin(|x_i - 1|), \quad i = 1, 2, \dots, n-1,$$

where $\Omega = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i \leq n, x_i \geq -1, i = 1, 2, \dots, n\}$.

Problem 8. [30] $F : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ defined by

$$F(x) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} x_1^3 \\ x_2^3 \\ 2x_3^3 \\ 2x_4^3 \end{pmatrix} + \begin{pmatrix} -10 \\ 1 \\ -3 \\ 0 \end{pmatrix}$$

where $\Omega = \{x \in \mathbb{R}^n : \sum_{i=1}^4 x_i = 3, x_i \geq 0, i = 1, 2, 3, 4\}$.

In the Figures 1–3, we adopt the performance profile by Dolan and Moré [8] to compare the performance of the MHSP method with that of PCG method and the PDY method based on: (1) the number of iterations, (2) CPU time and (3) number of functions evaluation. It can be easily seen that all the curves with respect to our method stay longer on the vertical axis which means our method MHSP outperforms the PCG and PDY methods with regards to numerical performance. Specifically, Figures 1–3 show that our method won about 94, 78 and 56 percent of the experiments in terms of number of iterations, CPU time(s) and number of functions evaluation respectively. Moreover, from the information reported in Table 2, it can be seen that our MHSP method performs better than the PCG and PDY methods in the experiments. In most of the experiments, MHSP method needs less number of iterations, CPU time(s) and the number of functions evaluation to obtain solution compared to the PCG and PDY methods. We can also see from Table 2 that, though the three solvers converge to exact solution of problem 4, only MSHP method converges to exact solution of problem 5 with initial points x_5, x_7, x_8 . Therefore, the experiments show that the MHSP method is very robust.

5. Applications in compressive sensing

5.1. Description of l_1 -norm regularization problem in compressive sensing

Compressive sensing is basically a procedure used for efficiently acquiring and reconstructing a signal. It compresses the signal being acquired at the time of sensing. Signals can have sparse or compressible representation either in original or some transformed domain. Problems arising from signal processing and statistical inference involve solving under-determined or ill-conditioned system of linear equations. A classical method consists of minimizing an objective function which contains a quadratic (l_2) error term combined with a sparseness-including (l_1) regularization term,

$$\min_x \frac{1}{2} \|y - Ax\|_2^2 + \tau \|x\|_1, \quad (5.1)$$

where $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$ ($m < n$) is a linear operator, $\tau > 0$ is a parameter, $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$, denotes l_1 -norm and Euclidean norm respectively, for $p = 1, 2$.

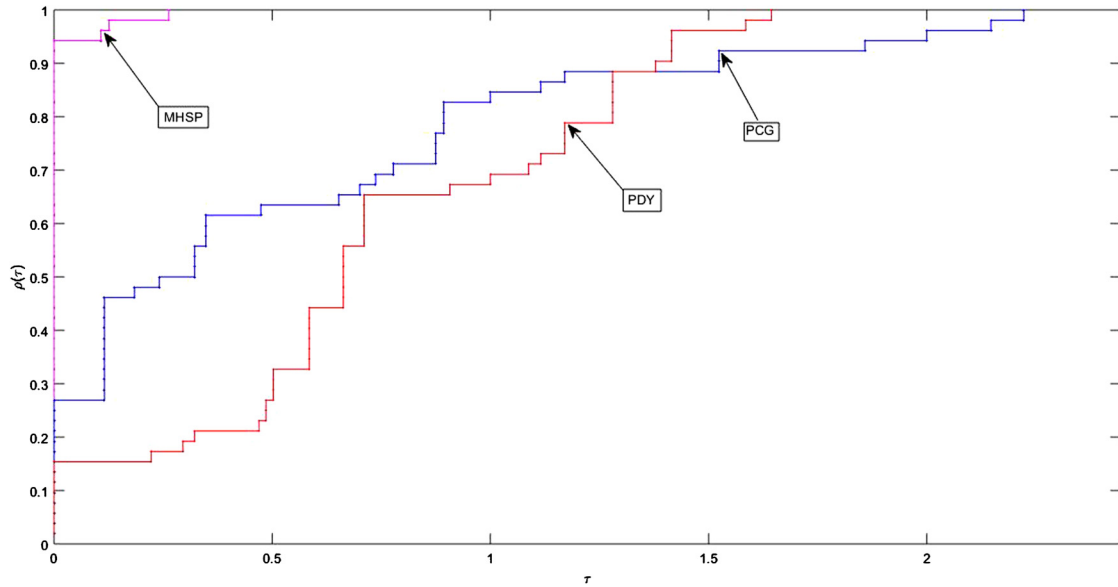


Fig. 1. Dolan and Moré performance profile with respect to number of iterations.

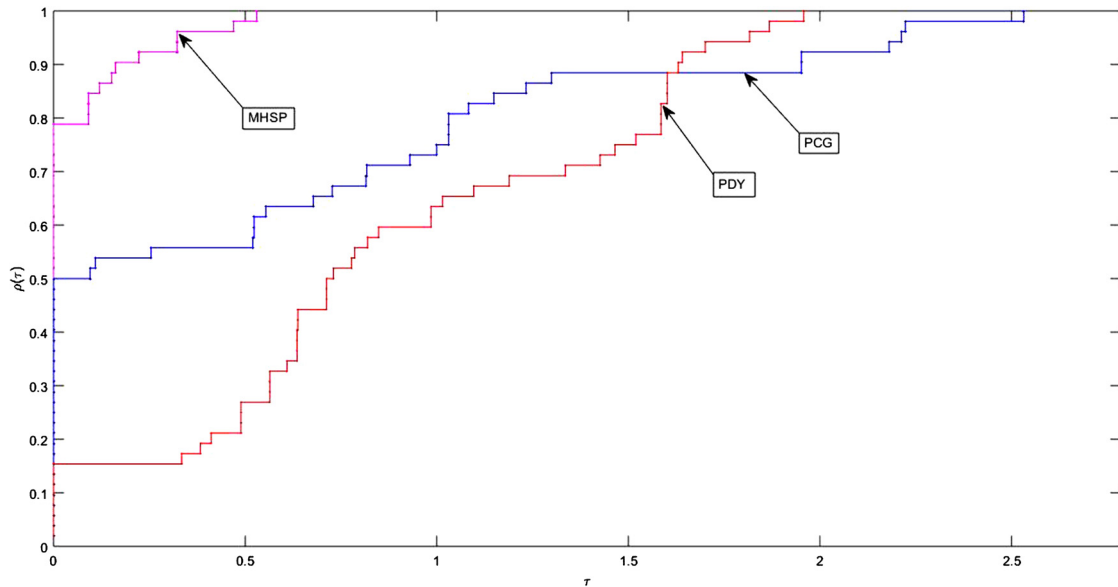


Fig. 2. Dolan and Moré performance profile with respect to number of functions evaluation.

One of many iterative methods for solving (5.1) is the method called gradient projection for sparse reconstruction (GPSR) proposed by Figueiredo et al. [9]. According to the work of Figueiredo et al. [9], any vector $x \in \mathbb{R}^n$ can be split into two terms i.e.

$$x = u - v, \quad u, v \geq 0, \quad u, v \in \mathbb{R}^n. \quad (5.2)$$

Let $(\cdot)_+$ denotes the positive-part operator defined as $(x)_+ = \max\{0, x\}$ then (5.2) satisfies $u_i = (x_i)_+$ and $v_i = (-x_i)_+$ for all $i = 1, 2, \dots, n$. Subsequently, we can represent the l_1 -norm of any vector as $\|x\|_1 = e_n^T u + e_n^T v$, where $e_n = (1, 1, \dots, 1)^T \in \mathbb{R}^n$. Therefore, the l_1 -norm problem (5.1) can be reformulated into the following

$$\min_{u, v} \frac{1}{2} \|y - A(u - v)\|_2^2 + \tau e_n^T u + \tau e_n^T v, \quad \text{s.t. } u \geq 0, \quad v \geq 0. \quad (5.3)$$

Furthermore, from [9], problem (5.3) can be easily rewritten as the quadratic program problem with box constraints as follows

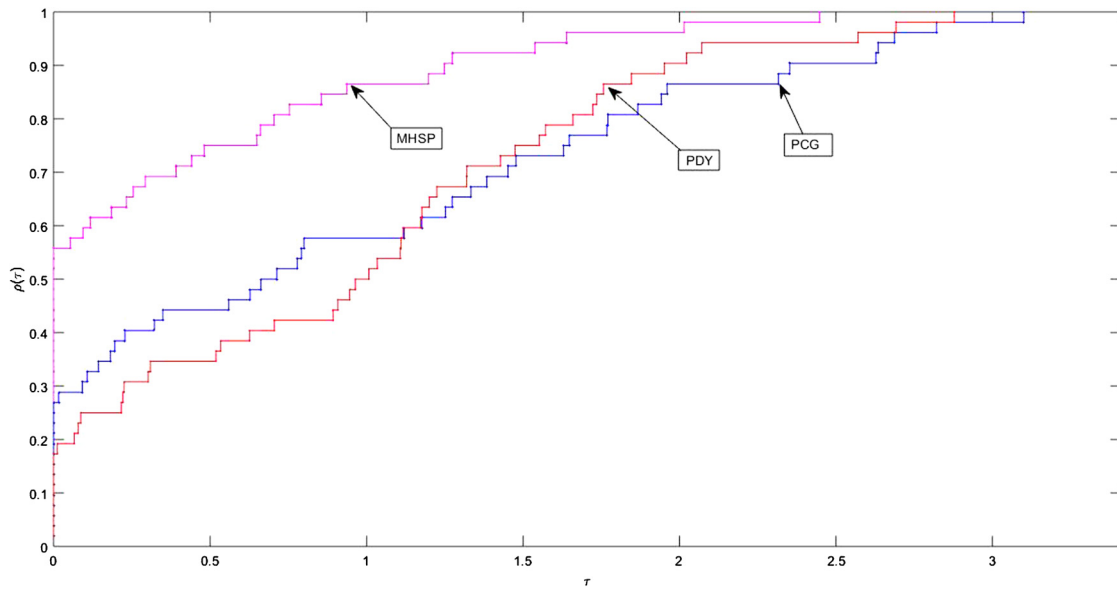


Fig. 3. Dolan and Moré performance profile with respect to CPU time.

$$\min_z \frac{1}{2} z^T B z + c^T z, \quad \text{s.t. } z \geq 0, \quad (5.4)$$

where

$$z = \begin{bmatrix} u \\ v \end{bmatrix}, \quad c = \tau e_{2n} + \begin{bmatrix} -b \\ b \end{bmatrix}, \quad b = A^T y, \quad B = \begin{bmatrix} A^T A & -A^T A \\ -A^T A & A^T A \end{bmatrix}. \quad (5.5)$$

It is not difficult to see that the matrix B is a positive semi-definite matrix.

In [28], Xiao et al. translated problem (5.4) into a linear variable inequality problem. They also indicated that a vector z is a solution of the linear complementary problem if and only if it is a solution of the following nonlinear equations

$$F(z) = \min\{z, Bz + c\} = 0, \quad (5.6)$$

where the function F is vector-valued and the “min” interpreted as component-wise minimum. It was shown in the work of Pang [22] and Xiao et al. [28] that the mapping F is Lipschitz continuous and monotone. Consequently, it can be solved by Algorithm 1 effectively. Next, we give the details of the numerical experiments.

5.2. Numerical experiments

In this subsection, we will demonstrate some experiments to show the performance of our proposed method. We consider the reconstruction of a sparse signal of length- n from m observations. The quality of restoration is assessed by the mean of squared error (MSE) to the original signal $\bar{x}$, i.e.,

$$MSE = \frac{1}{n} \|\bar{x} - x^*\|^2,$$

where x^* is the restored signal. In this experiment, the measurement y contains noise, $y = A\bar{x} + \omega$, where ω is the Gaussian noise distributed as $N(0, 10^{-4})$ and A is the Gaussian matrix generated by command `randn(m, n)`, in Matlab. The size of the signal is selected with $n = 2^{12}$ and $m = 2^{10}$. The original signal contains 2^7 randomly nonzero elements.

We compare our MHSP method with the recent PCG method [18] in order to show the efficiency of our method. The parameters used for both MHSP method and the PCG method are taking as $\kappa = 10$, $\sigma = 10^{-4}$ and $\rho = 0.5$, $r = \eta = 0.2$. We use $f(x) = \frac{1}{2} \|y - Ax\|_2^2 + \tau \|x\|_1$ as the merit function. We run each code with the same initial point, use the same continuation technique on parameter τ , and only observe the convergence behaviour of each method to obtain a solution with similar accuracy. We start the iteration by the measurement signal, i.e., $x_0 = A^T y$, and terminate when the relative change of the objective function satisfies

$$\frac{\|f(x_k) - f(x_{k-1})\|}{\|f(x_{k-1})\|} < 10^{-5}.$$

In order to have a comprehensive view of the performances of both the MHSP method and the PCG method, we plot four graphs (see Fig. 5) to demonstrate their convergence behaviours from the trends of MSE and the objective function values

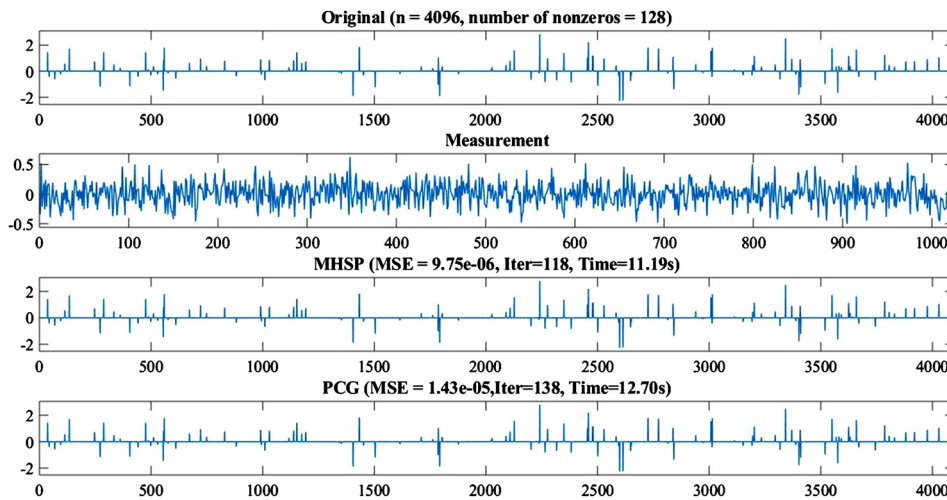


Fig. 4. From top to bottom: the original signal, the measurement, and the recovery signals by MHSP and PCG methods.

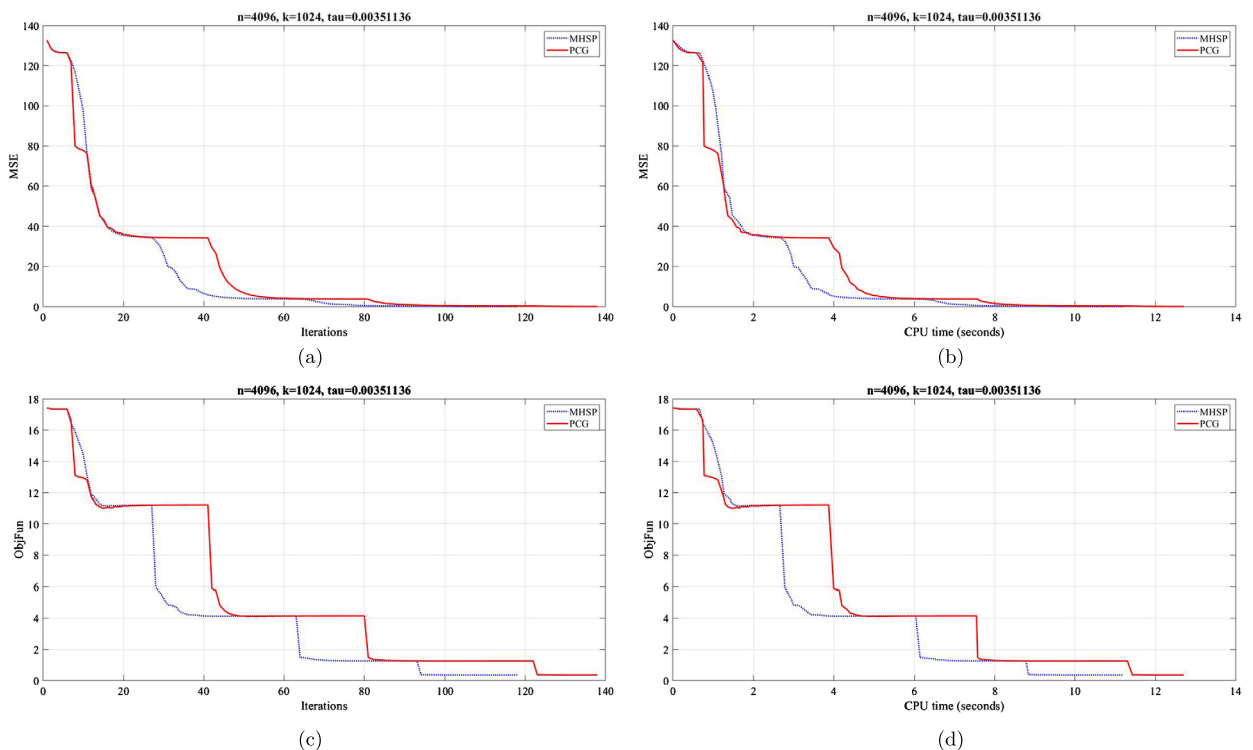


Fig. 5. Comparison results of MHSP and PCG methods. The horizontal axes of (a) and (c) represent number of iterations while that of (b) and (d) represent CPU time in seconds. The vertical axes of (a) and (b) represent the MSE while that of (c) and (d) represent the function values.

along with the number of iterations and CPU time increasing. From Fig. 4, we can see that the descent rates of MSE and objective function values of our proposed MHSP method are faster than that of the PCG method. Moreover, we repeated the experiments for different noise samples and the average of ten (10) is also computed. Table 3 shows that MHSP method is faster and needs less number of iterations than the PCG method.

6. Conclusions

In this paper, we proposed a modified Hestenes-Stiefel spectral conjugate gradient projection (MHSP) method for convex constrained monotone nonlinear equations with application in signal recovery. The search direction of our proposed algorithm is bounded and also satisfies sufficient descent condition. The proposed method inherits some nice properties of

Table 3

Ten experiment results with average result of ℓ_1 -norm regularization problem for MHSP and PCG methods.

	MHSP			PCG		
	MSE	ITER	CPU(s)	MSE	ITER	CPU(s)
	2.20E-05	125	12.11	2.39E-05	151	12.55
	1.56E-05	131	11.08	1.78E-05	148	12.31
	1.18E-05	130	12.36	1.81E-05	130	11.22
	2.23E-05	111	9.67	2.84E-05	125	10.64
	1.25E-05	127	10.50	2.03E-05	133	11.00
	9.75E-06	118	11.19	1.43E-05	138	12.70
	1.29E-05	116	9.55	1.09E-05	147	11.78
	3.35E-05	133	10.98	5.76E-05	134	10.55
	6.19E-05	97	8.94	2.35E-05	114	10.44
	1.92E-05	139	14.61	2.35E-05	140	12.50
Average	2.21E-05	122.7	11.099	2.38E-05	136	11.569

conjugate gradient methods such as derivative-free and low memory requirement which makes it applicable to solve monotone large-scale nonlinear equations with convex constraints. The global convergence under some suitable assumptions was established. In our numerical experiments, MHSP outperformed the PCG method [18] and the PDY method [17] in terms of number of iterations, CPU time and number of functions evaluation. In addition, we successfully applied our proposed MHSP method to deal with ℓ_1 -norm regularization problem in compressive sensing.

Acknowledgements

We wish to thank the associate editor and the two anonymous referees for their constructive comments and suggestions which greatly improved the manuscript. The authors acknowledge the financial support provided by King Mongkut's University of Technology Thonburi through the "KMUTT 55th Anniversary Commemorative Fund". The first author was supported by the Petchra Pra Jom Klao Doctoral Scholarship Academic for PhD. Program at KMUTT (Agreement No. 42/2560). This project was supported by the Theoretical and Computational Science (TaCS) Center under Computational and Applied Science for Smart Innovation Research Cluster (CLASSIC), Faculty of Science, KMUTT. Moreover, Poom Kumam was supported by the Thailand Research Fund and the King Mongkut's University of Technology Thonburi (KMUTT) under the TRF Research Scholar (Grant No. RSA6080047).

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